

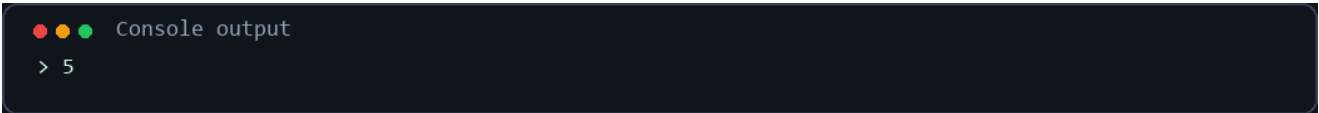
Java String Operations

15 separate Java programs with brief explanations and console-output screenshots.

2. Find String Length

The `length()` method returns the number of characters in a string, including spaces.

```
class StringLength {  
    public static void main(String[] args) {  
        String text = "Hello";  
        System.out.println(text.length());  
    }  
}
```

A terminal window with a dark background. It shows the text "Console output" in green, preceded by three colored dots (red, yellow, green). Below it, the output "> 5" is shown in white.

Console output
> 5

3. Concatenate Strings

The `+` operator joins two strings to form one longer string.

```
class ConcatenateString {  
    public static void main(String[] args) {  
        String first = "Hello";  
        String second = " World";  
        System.out.println(first + second);  
    }  
}
```

A terminal window with a dark background. It shows the text "Console output" in green, preceded by three colored dots (red, yellow, green). Below it, the output "> Hello World" is shown in white.

Console output
> Hello World

4. Compare Strings

`equals()` compares the text stored in two strings and returns true when they match.

```
class CompareString {  
    public static void main(String[] args) {  
        String a = "Java";  
        String b = "Java";  
        System.out.println(a.equals(b));  
    }  
}
```

A terminal window with a dark background. It shows the text "Console output" in green, preceded by three colored dots (red, yellow, green). Below it, the output "> true" is shown in white.

Console output
> true

5. Convert to Uppercase

`toUpperCase()` changes all lowercase letters in a string to uppercase letters.

```
class UppercaseString {  
    public static void main(String[] args) {  
        String text = "hello";  
        System.out.println(text.toUpperCase());  
    }  
}
```

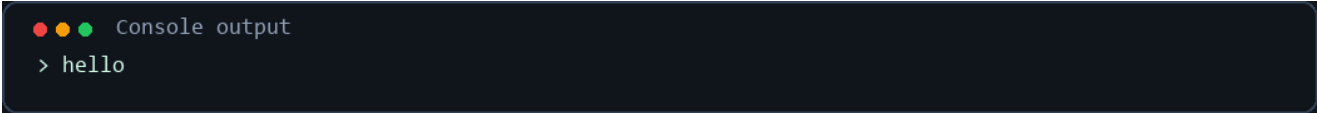
A terminal window with a dark background. It shows the text "Console output" in green, preceded by three colored dots (red, yellow, green). Below it, the output "> HELLO" is shown in white.

Console output
> HELLO

6. Convert to Lowercase

`toLowerCase()` changes all uppercase letters in a string to lowercase letters.

```
class LowercaseString {
    public static void main(String[] args) {
        String text = "HELLO";
        System.out.println(text.toLowerCase());
    }
}
```



Console output
> hello

7. Extract a Substring

`substring(start, end)` extracts characters from start up to, but not including, end.

```
class SubstringExample {
    public static void main(String[] args) {
        String text = "Programming";
        System.out.println(text.substring(0, 7));
    }
}
```



Console output
> Program

8. Find Character Position

`indexOf()` returns the first position of a character. Positions start from 0.

```
class IndexOfExample {
    public static void main(String[] args) {
        String text = "Hello";
        System.out.println(text.indexOf('o'));
    }
}
```



Console output
> 4

9. Replace Text

`replace()` changes every occurrence of the specified text with new text.

```
class ReplaceString {
    public static void main(String[] args) {
        String text = "Java is fun";
        System.out.println(text.replace("fun", "powerful"));
    }
}
```



Console output
> Java is powerful

10. Split a String

`split()` divides a string into an array using a separator. This program prints each array item.

```
class SplitString {
    public static void main(String[] args) {
        String text = "Java,Python,C++";
        String[] languages = text.split(",");

        for (String language : languages) {
            System.out.println(language);
        }
    }
}
```

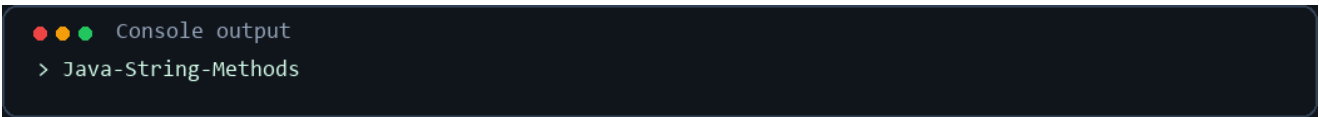


```
Console output
> Java
> Python
> C++
```

11. Join Strings

`String.join()` combines multiple strings by placing the chosen separator between them.

```
class JoinString {
    public static void main(String[] args) {
        String result = String.join("-", "Java", "String", "Methods");
        System.out.println(result);
    }
}
```

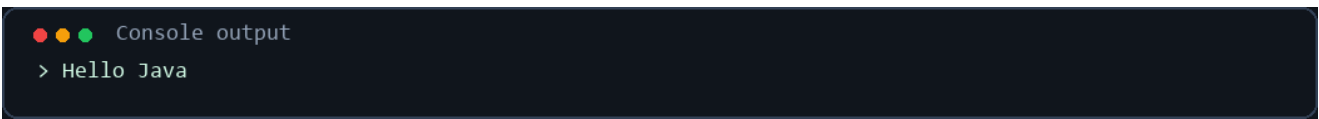


```
Console output
> Java-String-Methods
```

12. Trim Spaces

`trim()` removes whitespace from the beginning and end of a string.

```
class TrimString {
    public static void main(String[] args) {
        String text = " Hello Java ";
        System.out.println(text.trim());
    }
}
```



```
Console output
> Hello Java
```

13. Reverse a String

`StringBuilder.reverse()` reverses the characters. `toString()` converts the result back to a String.

```
class ReverseString {
    public static void main(String[] args) {
        String text = "Hello";
        String reversed = new StringBuilder(text).reverse().toString();
        System.out.println(reversed);
    }
}
```

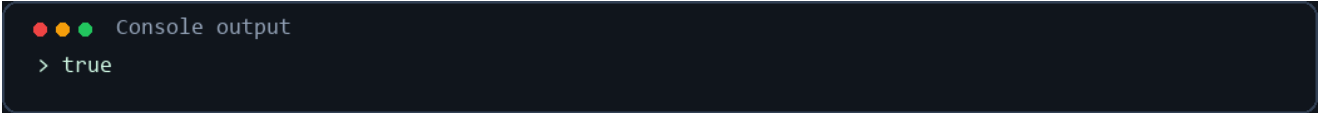


```
Console output
> olleH
```

14. Check Starts With

`startsWith()` checks whether a string begins with specified text and returns true or false.

```
class StartsWithString {  
    public static void main(String[] args) {  
        String text = "Hello Java";  
        System.out.println(text.startsWith("Hello"));  
    }  
}
```

A dark-themed terminal window with three colored dots (red, yellow, green) in the top left corner. The text "Console output" is displayed in a light gray font. Below it, the output "> true" is shown in a light blue font.

Console output
> true

15. Check Ends With

`endsWith()` checks whether a string ends with specified text and returns true or false.

```
class EndsWithString {  
    public static void main(String[] args) {  
        String text = "Hello Java";  
        System.out.println(text.endsWith("Java"));  
    }  
}
```

A dark-themed terminal window with three colored dots (red, yellow, green) in the top left corner. The text "Console output" is displayed in a light gray font. Below it, the output "> true" is shown in a light blue font.

Console output
> true

16. Check Whether a String Is Empty

`isEmpty()` returns true when a string contains zero characters.

```
class EmptyString {  
    public static void main(String[] args) {  
        String text = "";  
        System.out.println(text.isEmpty());  
    }  
}
```

A dark-themed terminal window with three colored dots (red, yellow, green) in the top left corner. The text "Console output" is displayed in a light gray font. Below it, the output "> true" is shown in a light blue font.

Console output
> true